Broadband Internet Expansion Motivates Primary Competition in U.S. Congressional Primaries

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Abstract

During the current nationalized era of American elections, conventional wisdom argues that members of Congress worry more about intraparty primary threats rather than general election out-party threats. If so, what mechanism explains the growth of intraparty primary threats to incumbent legislators? Leveraging the exogenous roll-out of broadband internet to congressional districts during the 2000s using a within-member design, we first find that growth in the information environment hastened greater partisan voting among geographic electorates at the expense of two-party competition. Congruently, broadband expansion motivated voluntary retirement and increased co-partisan primary opposition, with null effects for out-party competition, suggesting information environments facilitate primary rather than general election threats. In response, we find that incumbents react with more partisan, but not ideologically extreme, behavior, shifting away from district-centered representation. Despite these behavioral shifts, electoral outcomes remain unchanged, suggesting incumbents successfully neutralize threats through anticipatory adaptation for partisan polarization.

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> "They've monetized conservative politics... they've done it. And elected officials need to understand what their motivation is. They're just all about ratings and clicks and getting eyeballs on their sites... They're not going to thank you for doing good policy... they're going to tear that apart. They play to anger."
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> U.S. Rep. Charlie Dent (R-PA)

> "The bloggers who came after me—some of them were so full of hatred that it is just not good for our politics... so I worry that this victory yesterday by Ned Lamont, which was a narrow victory, will send a message across our state and our country that the Democratic party has taken – has been taken over by people who are not from the mainstream of America."
>
> U.S. Sen Joe Lieberman (D-CT)

Widely regarded as a moderate and one of the most bipartisan legislators in the U.S. House of Representatives, U.S. Congressman Charlie Dent (R-PA) lamented that party activists have "monetized conservative politics" not towards "doing good policy" but towards playing to "anger" among a partisan base.[1] Speaking to his observations in Congress, Rep. Dent implored elected officials to understand the motivation of online partisan media is to elicit "clicks and getting eyeballs on their sites" by deploying anger towards the opposing party rather than an attempt to galvanize support for bipartisan legislative solutions towards solving the nation's policy problems. In a similar vein and speaking on the heels of a narrow primary loss to businessman Ned Lamont in the 2006 Connecticut U.S. Senate Democratic primary, U.S. Senator Joe Lieberman (D-CT) accused online "bloggers" of being "full of hatred" during the primary campaign.[2] Speaking at further length in an interview with local media, Sen. Lieberman framed his narrow loss in the primary as a potential message that the Democratic Party "has been taken over by people who are not from the mainstream of America", implicitly arguing that online activists opposing his re-election were more partisan and ideologically rigid than those in the mainstream." Ultimately, Sen. Lieberman would

[1] PBS FRONTLINE Interview: Charlie Dent. https://www.pbs.org/wgbh/frontline/interview/charlie-dent-2/

[2] Lieberman's remarks following the 2006 Connecticut Democratic Senate primary. https://www.newsbusters.org/blogs/nb/mark-finkelstein/2006/08/09/lieberman-some-bloggers-bigoted-no-dem-can-dissuade-him-indie

be re-elected in the 2006 general election as an independent and would continue to caucus with Senate Democrats following his primary defeat.

The motivating quotes by Congressman Dent and Senator Lieberman highlight a fundamental tension, that members of Congress are increasingly facing credible primary challengers while competition declines among general election opponents (Boatright, 2013; Hirano & Snyder, 2019). While scholarly attention focusing on incentives provided by increasing primary competition affords incumbents (Brady, Han & Pope, 2007) and how primaries motivate partisan polarization in Congress (Hirano et al., 2010) is well documented, less is understood as to what mechanism explains the growth of intraparty primary threats to incumbents during a period of resurgent mass partisanship (Hetherington, 2001), competitive primaries (Boatright, 2013), and an overall decline in two-party competition in congressional elections. While recent work rightly notes that primaries are increasingly drawing more, and better-funded, candidates (Porter & Treul, 2024) and becoming more competitive contests (Boatright, 2013), we know relatively little about *why* primary challenges to incumbents are rising in prominence, a pattern widely noted in both the scholarly literature and politicians' own accounts. Moreover, there is no theory that reconciles this rise in primary challenge threats with the observed high likelihood of success by incumbents in primary elections (Treul et al., 2022), suggesting a lack of theory linking the rise of a partisan primary flank, in an increasingly nationalized and partisan landscape, to successful legislative adaptation by incumbents in response.

In this article we fill this gap in the literature by making a similar argument alluded to by Congressman Dent and Senator Lieberman, that changes in the information environment lead to a greater primary threat to members of Congress and, in response, to greater partisan-centered behavior by legislators designed to neutralize the emergence of a partisan threat in a primary contest. As such, the remainder of our paper is organized as follows. First, we articulate our theoretical framework resting on three key propositions: (1) that broadband expansion opened access to partisan media and nationalized political information (Lelkes, Sood & Iyengar, 2017), leading to more partisan constituencies; (2) broadband expansion increased voluntary retirement and competition to incumbents in primaries but not in the out-party across a series of measured outcomes, suggesting an intraparty mechanism linking broadband expansion and primaries; and (3) broadband expansion motivated legislators to engage in more partisan behavior on Capitol Hill

at the expense of district-focus representation, providing for a key mechanism towards neutralizing the electoral consequences of the changing partisan media landscape.

Second, we begin our analysis with finding robust evidence for the foundational assumption of our theory at multiple subnational levels, that broadband motivates more partisan constituencies (i.e., sorting constituencies towards partisan poles) at the expense of two-party competition. From there, we present a series of tests resulting from an identification strategy leveraging the as-if random exogenous assignment of broadband internet among members of Congress leveraging a new comprehensive dataset linking the geographic roll out of broadband internet, the nature of primary competition facing incumbents, and subsequent legislator response during the expansion of broadband internet during the 2000s. We find strong evidence that broadband expansion within-member facilitated a higher likelihood of voluntary retirement, lower likelihood of being unopposed, and a greater number of candidates emerging in co-partisan primaries. We find no evidence of these relationships in the opposing party's primary, providing evidence for our theoretical expectation of a co-partisan mechanism linking broadband and primary elections. Turning to representational outcomes we then find that broadband expansion facilitates partisan voting and more party focused rather than district focused legislative styles (Bernhard & Sulkin, 2018). Lastly, we find a null effect for broadband expansion on electoral success, suggesting incumbents successfully neutralize primary threats through anticipatory adaptation for partisan polarization.

Taken together, our results indicate that radical changes to the information environment provided by the expansion of broadband internet coincided with nationalized elections and greater primary threats to partisan primaries. Indeed, our work provides a theoretical account as to *why* members of Congress fear primary threats despite their high renomination rates and also speaks to a potential exogenous demand of partisan representation on Capitol Hill. Relatedly, our work concurs with emerging scholarship positing that not only are primaries becoming more crowded with the rise of amateur candidates over time (Porter & Treul, 2024; Algara et al., 2026), but that they are also contributing to increasing incentives for partisanship on Capitol Hill (Theriault, 2008).

1 Theoretical Framework: Broadband, Primary Threat, & Partisan Repositioning

1.1 Broadband, Information Costs, & Nationalized Partisan Electorates

The foundational premise of our theoretical model rests on the relationship between broadband internet and the partisan composition of geographic constituencies. The most direct evidence for this relationship comes from Lelkes, Sood & Iyengar (2017), who leverage the physical infrastructure of telephone lines as an instrument for broadband adoption and find that access to high-speed internet significantly increases affective partisan hostility among the American public. Their identification strategy exploits the fact that the pre-existing telephone network — built without reference to political considerations — determined where broadband providers could first deploy service, providing plausibly exogenous variation in access. The causal estimate is striking: broadband adoption produced measurable increases in partisan animus toward the opposing party, consistent with the theoretical expectation that cheaper access to partisan information and like-minded political communities hardens partisan identities among geographic electorates. We build directly on Lelkes, Sood & Iyengar's (2017) foundational finding, extending their demand-side evidence on mass partisan attitudes forward to the supply-side behavioral responses of incumbent legislators — a contribution their design cannot address.

The theoretical mechanism linking broadband to partisan constituency sorting operates through the *mobilization* of already-engaged partisans rather than the conversion of politically disengaged audiences. Zaller (1992) demonstrates that political information flows from elites to mass publics through communications media, with high-information citizens most susceptible to elite partisan cues. Prior (2007) shows that expanding media options concentrates political information consumption among the already-engaged, those citizens most likely to participate in primary elections rather than the general electorate at-large. Prior to broadband, monitoring an incumbent's voting record required substantial effort; broadband dramatically lowered these costs for the engaged partisan subset of the electorate, enabling real-time monitoring of incumbent behavior and facilitating the organizational infrastructure required to mount credible

primary challenges. Yamaya (2025) provides direct evidence for this cost-reduction logic in the campaign finance domain, demonstrating that broadband's effect on political engagement operated by lowering transaction costs for already-motivated partisan citizens rather than creating new political interest. The critical mechanism is thus not preference change but mobilization: cheaper information produces more organized primary voters, which in turn produces more credible co-partisan challenger recruitment.

The 2002 to 2008 window of our study coincided with the emergence of what Karpf (2012) documents as the partisan netroots, including organizations such as MoveOn.org, Daily Kos, and ActBlue, that organized explicitly around party accountability and primary challenger recruitment at the congressional district level. Beyond organizational infrastructure, broadband expanded access to partisan media across both ideological flanks simultaneously. DellaVigna & Kaplan (2007) provide the foundational causal study of partisan cable news showing that Fox News availability increased Republican vote share, while Martin & Yurukoglu (2017) document symmetric persuasion effects across partisan outlets. Stroud (2011) demonstrates that partisan selective exposure in online media concentrates among high-engagement citizens, precisely the primary voter most consequential for incumbent electoral security. Together, these supply-side information dynamics compound the mass partisan sorting documented by Lelkes, Sood & Iyengar (2017), generating our foundational assumption (A1): broadband expansion produces more partisan geographic constituencies as engaged partisans sort toward their partisan poles. Trussler (2021) provides further evidence of this dynamic at the district level, showing that broadband expansion reduced split-ticket voting and eroded the incumbency advantage as voters in high-information environments prioritized national partisan signals over local candidate evaluations — demand-side evidence that the information environment reshapes the effective partisan composition of the active primary electorate that incumbents must navigate.

1.2 Broadband, Strategic Retirement, & Primary Opposition

Greater partisan sorting among geographic constituencies raises the credibility of intra-party flanking threats faced by incumbent members of Congress. The seminal work of Jacobson (1983) on strategic candidate emergence establishes that challenger entry decisions are a function of anticipated competitive-

ness, with potential candidates more likely to enter when the probability of defeating the incumbent is higher. As broadband expands within a congressional district, the constituency becomes more partisan, and incumbents whose voting records deviate from the partisan preferences of their primary electorate face an increasingly credible flanking threat. For incumbents perceiving their re-election odds as insufficient in this more partisan environment, voluntary retirement emerges as the rational strategic response, consistent with Mayhew's (1974) model of single-minded reelection seekers who exit when the costs of competition outweigh the benefits of continued service (Thomsen, 2017). This generates our first hypothesis (H1): broadband expansion should increase the probability of voluntary retirement among incumbent members of Congress.

Beyond motivating strategic exit, a more partisan constituency should attract co-partisan primary challengers to incumbents seeking renomination. Boatright (2013) documents the rise of ideologically-motivated primary challenges to incumbents, while Bawn et al. (2012) reconceptualize parties as coalitions of intense policy demanders who use nominations to extract policy concessions from legislators. Broadband lowered the barriers to entry in primary campaigns for partisan activists lacking prior elected experience, candidates unconstrained by the strategic calculus facing experienced politicians whose electoral careers could be derailed by a primary loss against an incumbent (Jacobson, 1983). Experienced candidates weigh the opportunity costs of challenging an incumbent against the probability of success, a calculus less favorable to entry even in more partisan environments. This distinction generates separable expectations for the type of co-partisan challenger attracted by broadband expansion: we expect a positive effect on co-partisan primary opposition broadly (H2) driven specifically by amateur rather than experienced challengers (H2a, H2b), and reflected in greater fundraising by the co-partisan challenger field (H2c). The organizational infrastructure of the netroots amplified this dynamic. Cohen et al. (2008) argue that party insiders including activists, donors, and interest groups effectively control nominations by coordinating resources and signals; the broadband-enabled netroots organizations represent precisely this class of intense policy demanders (Bawn et al., 2012) now operating at the congressional district level, lowering the financial and organizational barriers for amateur co-partisans to contest incumbents they perceive as insufficiently loyal. Critically, since the mechanism is grounded entirely in co-partisan sorting

and intra-party accountability, out-party primary dynamics should be unaffected, providing a theoretically clean placebo test (H3).

1.3 The Primary Threat: Broadband & Partisan Legislator Behavior

Confronted with a more partisan constituency and a more credible flanking threat, rational legislators responding to the anticipated reactions of their primary electorate (Arnold, 1990) should adjust their legislative behavior toward greater partisan loyalty. The central empirical question is whether this behavioral response takes a *partisan* or *ideological* form, a distinction with important implications for understanding the mechanism linking information environments to legislative polarization.

Lee (2009) demonstrates that party unity in Congress is substantially driven by team loyalty and positional competition rather than ideological distance alone. A signal from a co-partisan primary challenger or partisan primary voter is fundamentally a loyalty signal — an incumbent votes with the opposing party too often — rather than a demand for ideological extremism. Conditional party government theory (Aldrich & Rohde, 2000; Rohde, 1991) reinforces this distinction: when parties are internally cohesive, rank-and-file members enforce loyalty norms through primary accountability rather than ideological purity tests. Under this framework, we expect broadband to increase party unity voting, the behavioral manifestation of partisan loyalty, while leaving ideological positioning as measured by DW-NOMINATE scores (Nokken & Poole, 2004) unchanged (H4, H5). The distinction between these two outcomes constitutes a critical mechanism test, separating the partisan accountability channel from a generic ideological extremism account. Trussler (2022) provides the most direct empirical precedent for this partisan mechanism, showing that legislators representing more informationally connected districts after the 2002 redistricting voted more in line with their party and the president — effects concentrated in party-line voting rather than ideological repositioning. Our paper extends this finding by identifying the primary threat channel through which the information environment generates behavioral shifts and demonstrating their anticipatory rather than reactive electoral character.

Beyond roll-call voting, the primary threat should redirect how members allocate their representational effort. Fenno's (1978) account of how members cultivate relationships with their geographic constituencies

identifies a fundamental tension between district-focused and party-focused representational styles. As the primary electorate becomes more partisan, incumbents face incentives to reallocate representational emphasis from constituency service toward partisan signaling (Grimmer, 2013; Park, 2023; Kistner et al., 2025). We operationalize this shift using the district-focused legislative style measure of Bernhard & Sulkin (2018), which captures whether a member's representational orientation emphasizes the district over party or policy considerations. We expect broadband expansion to reduce district-focused style as incumbents respond to the partisan primary threat (H6). However, this behavioral shift should manifest in representational orientation rather than material resource allocation: incumbents should not reduce legislative effectiveness, constituency staffing, or Members' Representational Allowance spending in response to a primary threat, as these resources serve constituents across the partisan spectrum and their withdrawal would be electorally visible (H7).

1.4 Neutralizing the Primary Threat: Successful Anticipatory Adaptation

The final prediction of our theoretical model concerns the electoral consequences of the behavioral adjustments documented in the preceding hypotheses. Arnold's (1990) anticipatory reactions framework establishes that legislators respond to the *threat* of primary accountability even before a challenger materializes, with incumbents who successfully reposition toward their primary electorate's preferences neutralizing the flanking threat before it produces an electoral consequence. Under this equilibrium, the behavioral adaptations in H4 through H7 are precisely what prevent electoral losses: incumbents who increase party unity and reduce district-focused style in response to broadband expansion are buying insurance against primary defeat.

This anticipatory adaptation prediction generates clean null expectations for incumbent electoral outcomes. If incumbents are successfully repositioning in response to the primary threat created by broadband expansion, we should observe no effect of broadband on the probability of winning renomination or the general election (H8). The absence of an electoral effect is not evidence against the mechanism but rather the predicted equilibrium outcome of successful behavioral adaptation. Mayhew (1974) establishes that single-minded reelection seekers update behavior to minimize the probability of electoral defeat; the

null electoral result confirms that this updating is occurring and is effective. This distinction matters for understanding the sources of legislative polarization more broadly. Theriault (2008) decomposes congressional polarization into member replacement and within-member behavioral change, finding that both processes contribute to rising partisan discipline. Our within-member design directly tests the behavioral change component, and the null electoral result confirms that incumbents are being disciplined rather than replaced — the behavioral adjustment we document is precisely what *prevents* the electoral turnover that would otherwise constitute the replacement channel. Taken together, our theoretical model predicts a three-stage causal chain: broadband expansion produces more partisan constituencies (A1), which motivates strategic exit and co-partisan primary opposition (H1–H3), which in turn disciplines incumbent behavior toward greater partisan loyalty and away from district-centered representation (H4–H7), while leaving electoral outcomes unchanged (H8).

Table 1: Summary of Testable Implications Derived from Theory

	Hypothesis	DV(s)	Expected Sign & Argument	
Foundational Assumption				
A1	Broadband increases constituency partisan deviation	Absolute partisan deviation	+	Broadband expands access to partisan media & nationalized political information, sorting constituencies toward their partisan poles
I. Hypothesized Outcomes				
H1	Broadband increases voluntary retirement	Voluntary retirement	+	Partisan sorting raises flanking threat; marginal incumbents exit strategically
H2	Broadband increases co-partisan primary opposition	Unopposed co-partisan primary	−	More partisan constituencies attract intra-party challengers
H2a	…driven by amateur challengers	Co-partisan amateur candidates	+	Broadband lowers entry barriers for partisan activists
H2b	…not experienced challengers	Co-partisan experienced candidates	NS	Career politicians' entry driven by other calculus
H2c	…reflected in co-partisan challenger fundraising	Co-partisan challenger (Log $)	+	Partisan threat draws financially resourced challengers into co-partisan primaries
H3	No effect on out-party competition (placebo)	Out-party primary outcomes; Out-party challenger (Log $)	NS	Mechanism is intra-party; out-party unaffected by co-partisan sorting
II. Mechanism Tests				
H4	Broadband increases party unity voting	Relative party unity	+	Preemptive alignment with party leadership to neutralize partisan flank
H5	No effect on ideological extremity	Ideological extremity (Nokken-Poole)	NS	Threat is partisan loyalty (procedural), not ideological (spatial)
H6	Broadband reduces district-focused style	District focused style	−	Partisan pressure redirects representational emphasis from constituency to party
H7	No effect on legislative capacity	LES, % constituency staff, MRA	NS	Behavioral shift is in orientation, not material resource allocation
III. Anticipatory Adaptation				
H8	No effect on incumbent electoral outcomes	Won primary, won general	NS	Successful behavioral repositioning neutralizes electoral consequences

NS = expectation of non-significant (zero) relationship between broadband & outcome.

2 Broadband & Partisan Sorting: Empirical Foundation

Having specified our core theoretical model, we turn to testing the core assumption that broadband expansion sharpens the partisan preference for subnational constituencies as a result of greater access to partisan media and nationalized political information that allows sorting of constituencies towards clarified partisan poles (Lelkes, Sood & Iyengar, 2017; Boxell, Gentzkow & Shapiro, 2022). One clear test of this assumption is to assess how local broadband expansion correlates with how much a constituency deviates from the national partisan baseline in a given presidential election, with higher values indicating a more partisan constituency. For example, in the recent 2024 presidential election, the two most partisan statewide electorates were found in Wyoming and West Virginia which voted about 25.38% and 20.59% more Republican than the country at-large[3]. Using this measure of partisan deviation from the national baseline, we expect greater broadband expansion within a subnational constituency (i.e., county, congressional district, state) to correlate with greater increases in relative partisan voting, reflecting more hardened partisan preferences among aggregate constituencies.

To test this assumption of our theoretical model, we measure broadband expansion at the county, congressional district, and state level aggregated from zip-code level data provided by the Federal Communication Commission (FCC), which we discuss in greater detail in section 3 specifying the research design testing our core hypotheses. This provides a test of the smallest unit of aggregation with corresponding electoral data, counties, and the largest being the state.[4] We specify a two-way fixed effects models with unit and year fixed effects with two parameters of interest, with one parameter specifying linear relationship between broadband expansion and partisan voting and the other parameter specifying a squared term specifying a potential non-linear relationship between expansion and voting. Allowing the functional form to take a linear and non-linear form, we allow the marginal effect of broadband on voting to vary with the level of broadband provisions (i.e., the number of providers). This approach allows for the estimation of the effect of broadband within counties, congressional districts, and states.

[3]Former President Trump won about 76.14% and 71.35% of the two-party vote in WY and WV while winning 50.75% of the two-party vote nationwide.

[4]While we have data at the ZIP-code level for broadband providers, the smallest unit at which election returns are reported is the precinct level, which may transcend multiple zip codes.

Table 2: Broadband Providers and Constituency Partisan Voting, 2000-2008

	U.S. Counties		U.S. Congressional Districts				U.S. States	
	(1)	(2)	(1)	(2)	(3)	(4)	(1)	(2)
Providers	0.542***	0.473***	0.774***	0.743***	0.746***	0.748***	1.165*	0.888
	(0.057)	(0.057)	(0.169)	(0.166)	(0.174)	(0.169)	(0.496)	(0.538)
Providers^2	-0.032***	-0.029***	-0.028***	-0.026***	-0.026***	-0.026***	-0.044	-0.027
	(0.003)	(0.003)	(0.006)	(0.006)	(0.006)	(0.006)	(0.024)	(0.024)
N	9336	9315	870	856	820	806	150	150
R^2	0.908	0.909	0.958	0.964	0.966	0.967	0.861	0.868
Adjusted R^2	0.862	0.863	0.915	0.926	0.928	0.929	0.785	0.784
Sample	Full	Full	Full	Full	Non-Redrawn	Non-Redrawn	Full	Full
Controls	-	✓	-	✓	-	✓	-	✓
Unit FE's	✓	✓	✓	✓	✓	✓	✓	✓
Time FE's	✓	✓	✓	✓	✓	✓	✓	✓

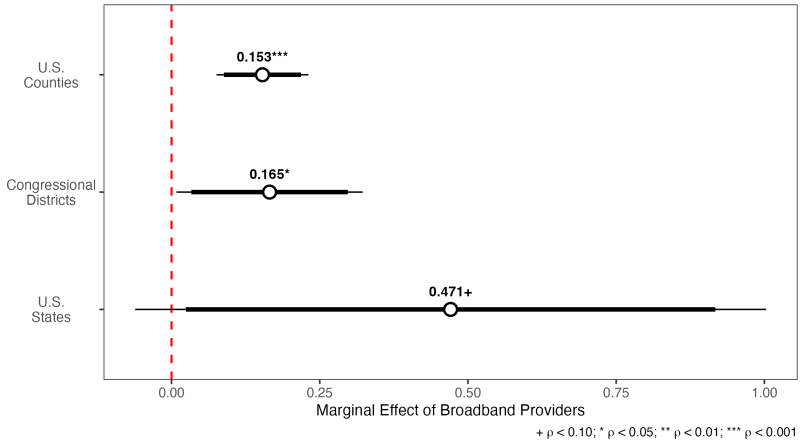
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2 presents results testing the core assumption of the theory using data from the 2000 to 2008 elections for counties and states, and from 2002 to 2008 for congressional districts given the 2002 redistricting cycle. The second model for each subnational unit includes a vector of time-varying demographic controls, such as median income, population density, racial composition, and the size of the foreign-born population. We also provide two models that incorporate a sample of congressional districts not drawn from the 2002–2007 cycles. Table 2 shows consistent evidence across both bivariate and full models that broadband expansion is significantly associated with partisan voting. However, the consistently negative and significant coefficient on the quadratic term indicates a nonlinear quadratic relationship between broadband expansion and partisan voting. Substantively, this suggests an inverted U-shaped relationship: at lower levels of broadband provision, expansion is associated with increases in partisan voting, but beyond a threshold, additional providers yield diminishing, and eventually negative, marginal effects.

To provide for an alternative view of this assumption test, we present the marginal effect of broadband expansion on partisan voting in Figure 1. Significant at $\rho < 0.05$, a one unit marginal effect increase in providers is associated with a 0.15% and 0.17% increase in partisan deviation from the national baseline at the county and district level, respectively. In substantive terms, that means a one standard deviation increase of providers at the county (σ =3.20) and district level (σ =3.65) is associated with about a 0.48% and 0.621% increase in partisan voting, a sizable shift from one election to the next during an era of nationalized

elections (Jacobson, 2015; Algara & Bae, 2024). With respect to states and perhaps owing to the far smaller sample size, the relationship between providers and partisan voting is significant at $\rho < 0.10$ with an increase of one provider being associated with a 0.47% increase in partisan voting. In substantive terms, a standard deviation increase in providers (σ =2.66) is associated with a 1.25% increase in partisan deviation from the national baseline. Taken together, our analysis finds strong support at each subnational level that broadband expansion is associated with a growing partisan electorate, pushing constituencies to their partisan poles during an era of nationalized elections. Congruently, this assumption shows that most subnational electoral arenas are becoming nationalized as a function of broadband expansion, suggesting that the national environment is aiding in the general lack of two-party competition across these arenas.

Figure 1: Broadband Expansion & Greater Partisan Voting in Geographic Electorates



Note: Estimates taken from fully specified models with all units. Congressional district marginal effect taken from full sample. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using unit clustered robust standard errors. The marginal effect of one additional broadband provider on partisan deviation from OLS regression models, holding everything else constant (i.e., ∂ Partisan Deviation/∂ Providers), is calculated using both additive and squared term such that ∂ Partisan Deviation/∂ Providers $= \hat{\beta}_1 + 2\hat{\beta}_2 \cdot$ Providers.

3 Research Design: Estimating the Broadband Effect

3.1 Data & Measurement

Having tested the core assumption that greater broadband expansion motivates partisan constituencies, we turn to articulating the core research design estimating the effect of broadband expansion across various outcome variables. Used in previous studies assessing the impact of broadband internet providers on electoral and legislative behavior (i.e., Trussler, 2021, 2022), we collect data from the FCC Form 477 that provides data on the number of broadband internet service providers (ISPs) at the smallest level of aggregation, the zip code level. Under Section 706 of the Telecommunications Act of 1996, the FCC collects standardized information from qualified broadband providers on whether they have one or more lines in service in each zip code. Beginning in 1999 and continuing until 2008, the Section 706 required the FCC to collect data on all ISPs that provide access service to consumers at speeds exceeding 200 kbps in at least one direction with respect to uploading or downloading speeds. These forms were aggregated and publicly released twice a year in June and December of every given year.

As we discussed at greater detail in the appendix, we construct our treatment measure using the following procedure. First, we collect all individual Form 477 filings from 2002 to 2008 and take the summer filing as our pre-election measure of broadband internet service providers. We begin with 2002 since that was the first election cycle featuring newly redistricted congressional districts provided by the 2000 U.S. Census. From there, we take this zip-code level data and merge it with a population-weighted cross-walk mapping zip codes to U.S. congressional districts. This provides us with the data structure needed to calculate the weighted average number of broadband providers in congressional district d during election t from zip codes indexed by z:

$$\text{providers}_{d,t} = \frac{\sum_{z \in d} \text{providers}_{z,t} \times \text{afact}_z \times \text{pop}_z}{\sum_{z \in cd} \text{afact}_z \times \text{pop}_z} \tag{1}$$

where providers$_{z,t}$ indicates the whole number of providers in a given zip code z at time t residing within congressional district d ($z \in cd$); afact$_z$ is the fraction of zip code z's population residing within congressional district d ($z \in cd$) taken from the cross-walk; and pop$_z$ is the population estimate residing within zip code z during the 2000 U.S. Census. This weighted average of broadband providers across the zip codes comprising each congressional district serves as our treatment variable, which given the weighting procedure is not restricted to integer values.

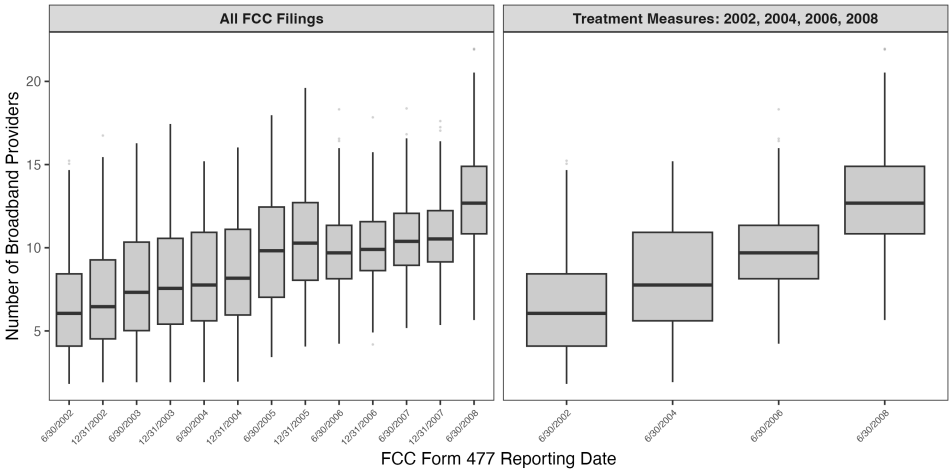
Figure 2 shows the over time and pooled distribution of the weighted average number of broadband providers within congressional districts from 2002 to 2008. As one can see, there is a great deal of variation on the number of broadband providers over time and between congressional districts Indeed, in a given year, the median number of providers within a congressional district was 9.3, with Figure 2A showing pronounced growth in the number of providers over time from 2002 to 2008. This measure of broadband providers for each congressional district serves as our treatment measure from the 2002 redistricting cycle to 2008.[5]

Now armed with our treatment variable at the year-congressional district unit of analysis, we merge this treatment with a given member of Congress serving from 2002 to 2008, providing for a member-level panel structure containing the specific treatment variable. From there, we add to the structure of these data by collecting three broad sets of outcome variables shown in the descriptive statistics of outcome variables below in Table 3. To begin, we collect three binary outcomes to assess strategic retirement and whether a legislator saw no candidate opposition within a co-partisan primary or the out-party's primary. On average, about 8% of legislators voluntarily retire from Congress in a given election cycle while 75% and 2% of co-partisan and out-party primaries are unopposed when an incumbent seeks re-election. Turning to the candidate primary opposition outcomes, we leverage data from Porter & Treul (2024) to collect the number of candidates that oppose an incumbent in a co-partisan primary and in the out-party's primary. We also leverage two count measures distinguishing between the number of experienced and amateur candidates opposing an incumbent's re-election bid in both primaries given the differing expectations of

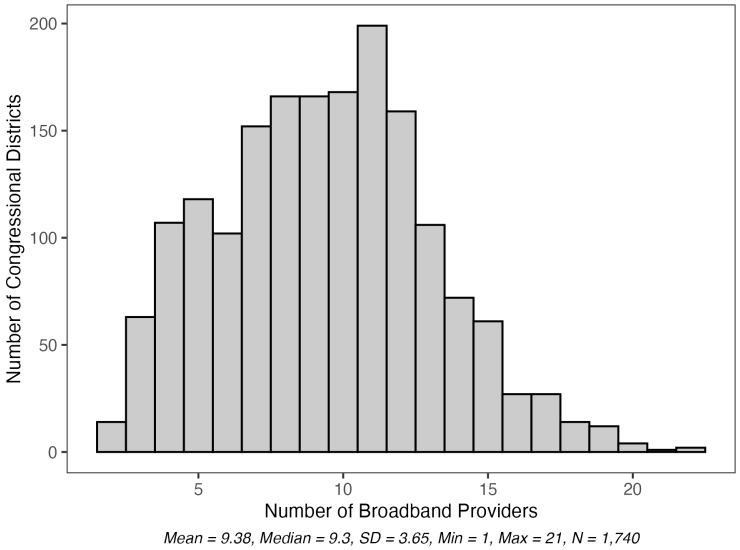
[5]We note that 2008 was the last year of data for the Form 477 being reported at the zip code level. While post-2008 data is available at the census tract level, we elect to leverage the entire 2002-2008 period given the consistency of the data collection of broadband providers for this time period.

Figure 2: Distribution of Broadband Providers in Congressional Districts, 2002-2008

(a) Overtime Distribution



(b) Pooled Distribution



Note: Histogram shows the distribution broadband internet service (ISP) providers within congressional districts for the 107th to the 110th Congresses encompassing the years 2002 to 2008. Data compiled from the Federal Communication Commission (FCC) Form 477 provider data by zip-code aggregated to congressional districts.

H2a and H2b with respect to co-partisan primary competition. Moreover, we also measure the logged fundraising total raised by the field of challengers in the co-partisan primary contesting the renomination of an incumbent and in the out-party's primary. These binary strategic calculus, count candidate pipelines, and campaign fundraising dependent variables serve as the outcomes of emphasis in the hypothesized

outcomes in Table 1, with the outcomes measuring opposition in the out-party primary serving as the salient placebo test derived from theory.

Moving down Table 3, we compile a broad array of data from a variety of sources (i.e., Nokken & Poole, 2004; Volden & Wiseman, 2014; Bernhard & Sulkin, 2018; Algara & Johnston, 2022; Treul et al., 2022; Crosson & Kaslovsky, 2025) that measures outcome variables measuring two types of broad representational outcomes, with the first set of outcome variables measuring partisan representation and the other measuring district-centered representation. Towards measuring partisan representational outcomes, we collect data on party unity, ideological extremity, the proportion of bills the member co-sponsored where the lead sponsor was from the same party, and the count of co-partisan co-sponsorships detailing the number of bills the member co-sponsored where the lead sponsor was from the same party. We center the outcome partisan-centered variables by the party median, with higher values signaling a greater degree of partisan representation.[6] We measure district-centered behaviors in the form of legislative effectiveness, the percentage of staff dedicated to constituency services centered by the chamber median, proportion of a legislator's Members' Representational Allowance (MRA) dedicated to constituency service centered by the chamber median, and three binary outcomes measuring if a member's legislative style is district-centered relative to party or policy focused (Bernhard & Sulkin, 2018). Congruently, higher values of these variables indicate a higher level of district-focused legislative behavior. Lastly, we have two outcome variables to measure the anticipatory adaptation expectations assessing if a legislator wins renomination or the general election in November. As one can see, re-nomination and re-election are high conditional on seeking re-election with 99.2% of incumbents that run in a primary successfully being renominated and 96.7% of incumbents successfully being re-elected.

[6]The descriptive statistics show the values for the panel of observations that encompass legislator seeking re-election, while the variable centering is done on the full partisan conference, which includes lame duck legislators not seeking re-election.

Table 3: Descriptive Statistics of Outcome Variables, 2002-2008

	N	Mean	Median	SD	Min	Max
I. Hypothesized Outcome Variables						
Strategic Retirement & Competition Outcomes						
Voluntary Retirement	1728	0.082	0	0.275	0	1
Unopposed Co-Partisan Primary	1580	0.751	1	0.432	0	1
Unopposed Out-Party Primary	1319	0.02	0	0.139	0	1
Candidate Primary Opposition Outcomes						
Co-Partisan Candidates Count	1728	0.647	0	1.429	0	15
Co-Partisan Amateur Candidates Count	1728	0.438	0	0.958	0	12
Co-Partisan Experienced Candidates Count	1728	0.209	0	0.678	0	8
Co-Partisan Challengers Primary Log Fundraising	1589	3.175	5.836	0	16.330	
Out-Party Candidates Count	1468	1.644	1	1.136	0	10
Out-Party Amateur Candidates Count	1468	1.364	1	1.048	0	7
Out-Party Experienced Candidates Count	1468	0.28	0	0.551	0	5
Out-Party Challengers Primary Log Fundraising	1589	0.140	1.354	0	15.171	
II. Mechanism Tests Outcome Variables						
Partisan Representation Outcomes						
Relative Party Unity	1499	-0.025	0.001	0.073	-0.446	0.065
Ideological Extremity (Nokken-Poole 1st Dim.)	1498	0.111	0.095	0.088	0	0.572
Proportion Co-Partisan Sponsors	1119	-0.022	-0.005	0.158	-0.867	0.391
Number of Co-Partisan Sponsors	1119	70.696	14	227.112	-311	1799
District Representation Outcomes						
Legislative Effectiveness	1466	1.905	2	0.752	1	3
% Constituency Staff	732	-0.109	0	9.795	-30.101	27.857
Constituency Spending (MRA)	732	0.01	0.007	0.076	-0.198	0.316
District vs. Party/Policy Focused Style	1740	0.158	0	0.365	0	1
District vs. Party Focused Style	623	0.441	0	0.497	0	1
District vs. Policy Focused Style	768	0.358	0	0.480	0	1
III. Anticipatory Adaptation Outcome Variables						
Incumbent Electoral Outcomes						
Won Primary Nomination	1589	0.992	1	0.09	0	1
Won General Election	1569	0.967	1	0.179	0	1

3.2 Identification Strategy

Now that we defined our treatment and outcome variables of interest, we turn to the identification strategy towards estimating the effect of broadband expansion on primary competition, legislative behavior, and anticipatory adaptation. To begin, we focus on the estimation strategy for the hypothesized outcomes

below in equation 2.

$$Y_{it} = \beta\, \textit{Providers}_{it} + \boldsymbol{\delta}\mathbf{X}'_{it} + \alpha_i + \gamma_{t\times p} + \varepsilon_{it} \tag{2}$$

where Y_{it} is outcome variable Y observed for legislator i during election t; $\textit{Providers}_{it}$ is the treatment variable of interest for legislator i at election t; $\mathbf{X}'_{it}$ is a vector of time-varying potential confounders such as seniority and electoral security for legislator i at time t; α_i are legislator fixed-effects accounting for time-invariant legislator characteristics that correlate with the voluntary retirement, the presence of competition, and degree of attracted primary competition; $t \times p$ are time fixed-effects interacted by party to account for partisan-year electoral shocks; and lastly an ε_{it} error term. β captures our estimand of interest, which is the effect of broadband providers on electoral outcomes within a legislator's electoral prospects. As such, equation 2 identifies variation in the outcome variable within-member and across cycle changes in broadband internet expansion, with β identifying the effect of an additional broadband provider on the outcome variable for a given legislator i. For the representational outcomes premised on a legislator reaction following broadband expansion, we specify equation 3.

$$Y_{i,t+1} = \beta\, \textit{Providers}_{it} + \mathbf{X}'_{it}\boldsymbol{\delta} + \alpha_i + \gamma_{t\times p} + \varepsilon_{it} \tag{3}$$

Note that the only substantive difference between both equations is that in equation 3 the outcome representational variables are observed in the subsequent Congress following the election t for $t+1$ to serve as a reaction to the expansion of broadband internet among legislator i's geographic constituency. While we specify each equation with legislator fixed-effects given that strategic retirement, representational outcomes, and the nature of the campaign once a legislator decides to seek re-election are ultimately *legislator-level* decisions, we present a specification of both equations with district fixed-effects in the appendix and yield results consistent with the member fixed-effects framework. Legislator fixed-effects further allow us to account for time-invariant unobserved heterogeneity in legislator sorting to districts, as identification in both equations rests on within-legislator variation in broadband expansion over time, net of legislator, year, and party fixed-effects. Across both equations we estimate uncertainty around our parameters clustered at the legislator level.

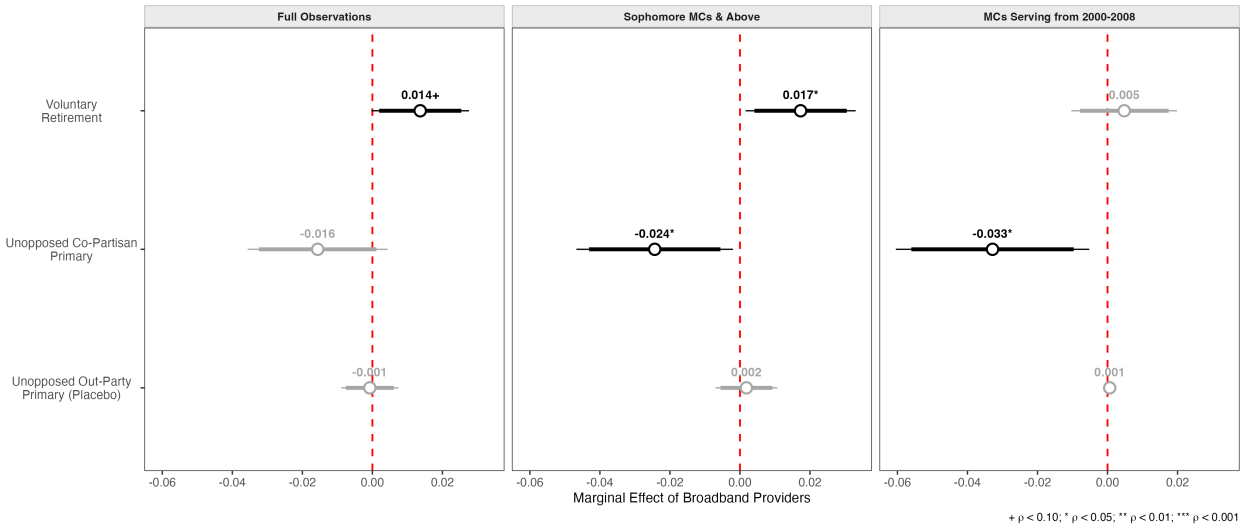
The underlying assumption in our identification strategy shown in both equations rests on the as-if random exogenous assignment of broadband internet among members of Congress. While the spatial distribution of broadband internet expansion was predicated on geographic considerations such as rural density and affluence of a given locality with respect to market incentives, we contend that the roll out of broadband was exogenous to political considerations and electoral competition. To assess the viability of this assumption, we conduct three empirical tests highlighted in greater detail in the appendix. First, we regress the number of broadband providers within a congressional district on a battery of political variables, including district partisan composition, incumbent vote share, legislative effectiveness, committee power, and freshman status, conditioning on district and year fixed effects. The null results across all political predictors indicate that broadband expansion was not systematically driven by the electoral or political conditions of a given district, consistent with the assumption that broadband ISPs responded to geographic and market considerations rather than political ones. Second, we estimate a leads-and-lags specification that jointly models the effect of broadband providers at $t-1$, t, and $t+1$ on our outcomes of interest. The null lead coefficient (β_{+1}) rules out anticipatory behavior and reverse causation, while the null lag coefficient (β_{-1}) rules out pre-existing political trends that would confound our estimates. Third, we estimate placebo models using general election vote share and electoral outcomes as dependent variables, as well as out-party primary opposition, which our theory does not predict broadband to affect. Taken together, the null results across these placebo outcomes confirm that broadband is not simply capturing pre-existing variation in electoral competitiveness and that the estimated effects are specific to the co-partisan electoral threat mechanism we theorize.

4 Results

4.1 Strategic Exit & Primary Opposition

We begin by evaluating our core hypotheses assessing if broadband expansion informs voluntary retirement among incumbents and primary opposition. Each forthcoming hypothesis test will be specified on three distinct samples given the panel structure of the data with the first including all legislator observations

Figure 3: Marginal Effect of Broadband on Strategic Retirement & Primary Opposition



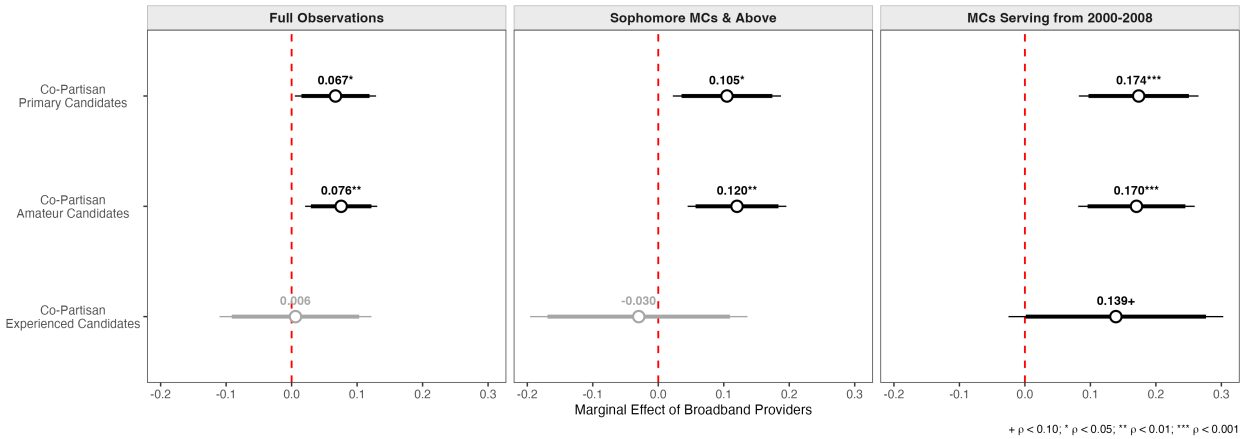
Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from linear probability models specified with vector of control variables, year × party fixed-effects, and member fixed-effects. Out-party primary opposition outcome serves as a placebo test.

in the panel, the second including only sophomore legislators or above that appear in the panel at least twice, and the third being a restrictive panel of legislators that served during the entire period of the panel, thus each legislator being observed the same number of times in the data for a total of four observations (i.e., being observed in 2002, 2004, 2006, & 2008).

Figure 3 presents results of a linear probability model predicting voluntary retirement and opposition to incumbents at the primary electoral stage. To begin, we find a consistent effect of broadband expansion on voluntary retirement among full observations and the sophomore and above panel, in line with the expectations of H1 with one additional broadband provider raising the probability of voluntarily retirement among both samples by 1.4% and 1.7%. We find no evidence of a significant effect among the restrictive sample of legislators serving from 2002 to 2008, which is not surprising given the higher service longevity of these legislators and lack of strategic retirement among this cohort. Congruently, we find a significant negative broadband effect on a lack of competition in an incumbent's co-partisan primary among the sophomore and above sample in addition to the restrictive sample with an additional provider lowering

the probability of an unopposed primary by 2.24% and 3.33%, respectively. While the effect is in the correct negative direction for the full set of observations, the relationship is insignificant when including legislators that only appear once in the data (i.e., freshman), suggesting consistent evidence for H2. Lastly, Figure 3 finds strong support for the theorized null relationship between broadband expansion and the probability of seeing unopposed opposition to an incumbent emerging in the opposing party's primary. Consistent with the placebo test expectation of H3, we find no evidence that broadband expansion drives out-party competition across all three samples.

Figure 4: Marginal Effect of Broadband on Co-Partisan Primary Opposition Candidate Pipeline



Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from poisson regression models specified with vector of control variables, year × party fixed-effects, and member fixed-effects.

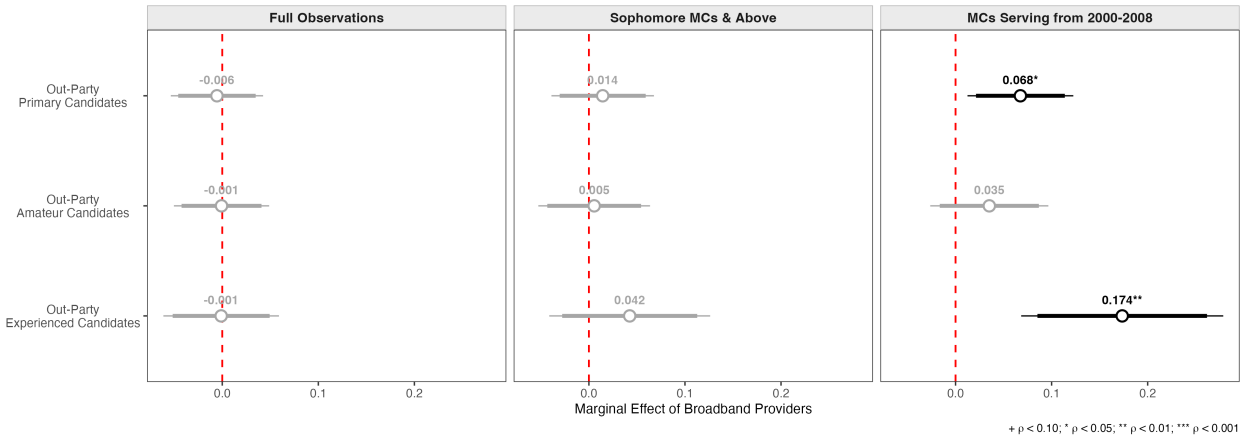
Moving on to the evaluation of H2a, H2b, and H2c we turn to poisson regression models predicting our challenger pipeline count variables assessing the degree of primary competition to legislators. The results of this analysis are conveyed in the marginal effects presented in Figure 4. We find strong evidence of a consistent positive effect of broadband expansion on the number of co-partisan primary challengers opposing a legislator's re-election across all three samples. Turning attention to the top row of Figure 4, we find that each additional broadband provider among a legislator's constituency raises the predicted number of primary challengers by 0.067, 0.106, and 0.174 for the three panel samples, respectively. These are sizable effects, as going from the minimum to the maximum value of broadband providers raises the

predicted count of challengers opposing an incumbent's renomination in the primary by 1.34, 2.10, and 3.48 candidates across the samples. Consistent with H2a and H2b, we find that this increase in candidate opposition to incumbents is driven by a rise of amateur challengers rather than experienced politicians hoping to continue their elected careers in the U.S. House. As we theorize, we posit that broadband expansion lowers the barriers to entry in the primary campaign featuring incumbents for partisan activists without elected experience (i.e., amateur candidates) that are unconstrained by the strategic calculus facing experienced candidates fearful that a primary loss to a co-partisan incumbent could derail electoral careers that are costly to cultivate (Jacobson & Kernell, 1983; Porter & Treul, 2024). Figure 4 find strong support for this proposition, showing a consistent positive broadband effect of 0.076, 0.120, and 0.170 on co-partisan amateur opposition to incumbents while this effect is null at standard levels of significance for co-partisan experienced candidates. This suggests that broadband expansion motivates a growing co-partisan challenger pipeline to incumbent re-election bids, but only among co-partisan amateurs and not experienced co-partisan politicians. To further support this finding, we turn to Figure 5 articulating a parallel analysis as Figure 4 but specified for our out-party placebo outcomes. Recall that H3 argues that broadband expansion should have no effect on out-party competition given that the mechanism linking broadband and primary competition is grounded in co-partisan sorting provided by an increasingly nationalized and partisan electorate faced by members of Congress. Figure 5 finds consistent null effects of broadband on the out-party primary candidate pipeline, further providing evidence that broadband expansion motivates only co-partisan opposition to a legislator rather than out-party opposition.[7]

Turning to evaluation of H2c and H3 with respect to challenger fundraising, Figure 6 shows the marginal effect of broadband providers on the logged inflation adjusted dollars raised by the collective challenging field to an incumbent legislator. The expectation is that broadband lowers the cost of prospective challengers to incumbents in a co-partisan primary and thus broadband should have a positive effect on the level of fundraising opposing an incumbent in the primary while this same effect should be null with respect to the fundraising of out-party challengers. As Figure 6 shows, we find evidence of a positive

[7]We note only two exceptions with the finding of a significant broadband effect among the full panel for out-party primary candidates and out-party experienced candidates. Given the lack of consistent evidence across outcomes and sample, we contend that these results may be spurious or unique to legislators that successfully survived electorally during the entire time period under our study.

Figure 5: Marginal Effect of Broadband on Out-Party Primary Opposition Candidate Pipeline



Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from poisson regression models specified with vector of control variables, year × party fixed-effects, and member fixed-effects. Out-party primary opposition outcome serves as a placebo test.

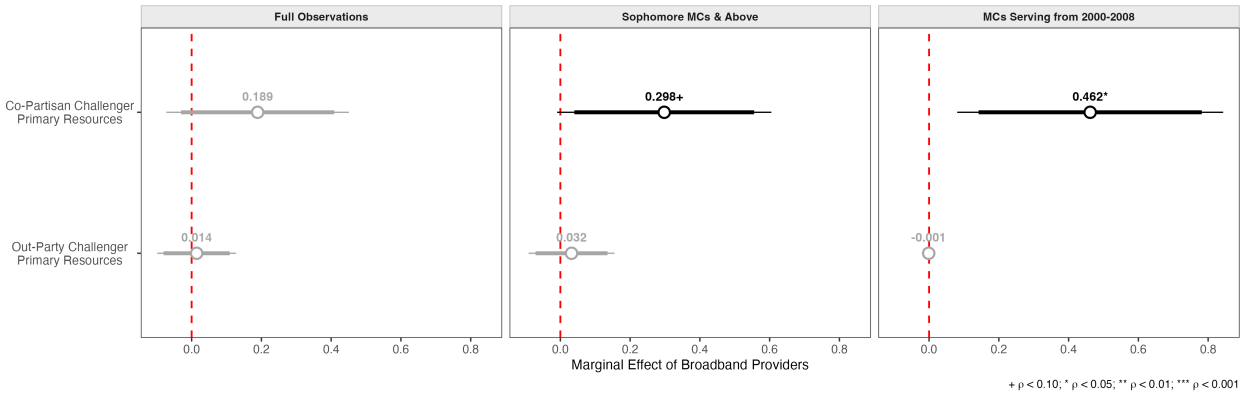
broadband effect on the fundraising of the co-partisan challenger among the sophomore panel and full panel. For each additional broadband provider, co-partisan challengers raised about 0.298 and 0.462 logged dollars in the sophomore and full panel. By contrast, we find clear null results for the opposing party, suggesting that broadband internet only motivates co-partisan electorates in the electoral arena rather than in the opposing party.

4.2 Mechanism Tests: Partisan vs. District Representation

We now turn to the mechanism tests evaluating whether broadband expansion shapes the partisan and district-centered representational behavior of legislators who choose to seek re-election. Figures Figure 7 and Figure 8 present the marginal effects of one additional broadband provider on our partisan and district representational outcomes, respectively, across all three panel samples.

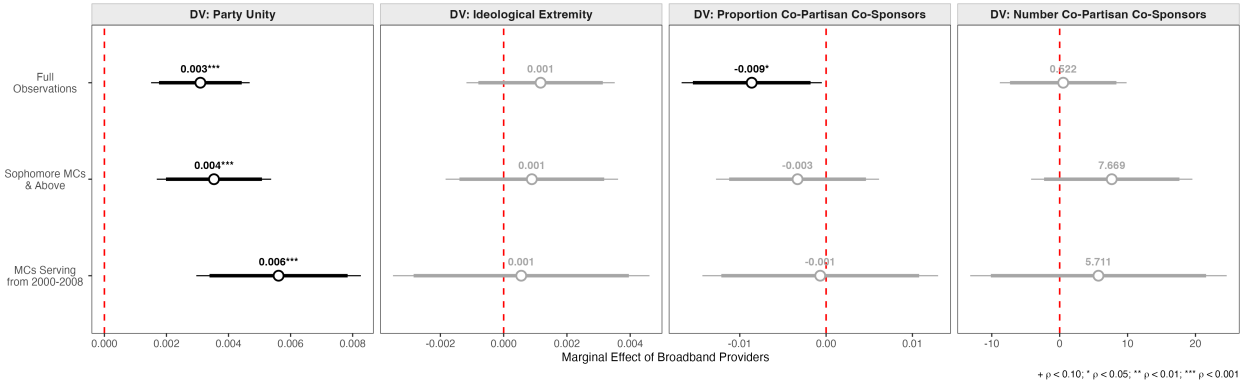
Figure 7 presents strong and consistent evidence in support of H4 and H5, the core mechanism test distinguishing partisan from ideological behavioral responses to broadband expansion. Across all three panel samples, we find a highly significant positive effect of broadband expansion on party unity

Figure 6: Marginal Effect of Broadband on Primary Challenger Fundraising Dollars



Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from OLS regression models specified with vector of control variables, year × party fixed-effects, and member fixed-effects. Out-party primary fundraising outcome serves as a placebo test.

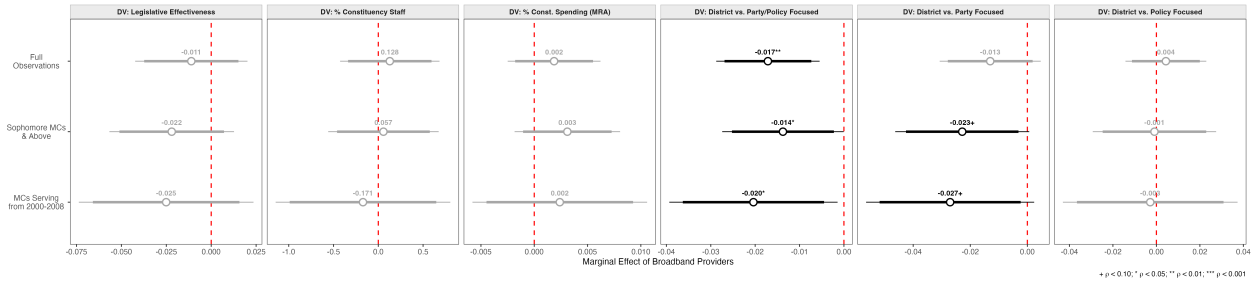
Figure 7: Marginal Effect of Broadband on Partisan Representational Outcomes



Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from OLS regression models specified with vector of control variables, year × party fixed-effects, and member fixed-effects.

voting, with one additional broadband provider increasing relative party unity by 0.003, 0.004, and 0.006 party unity score points for the full, sophomore and above, and complete panel samples, respectively. In substantive terms, a one standard deviation increase in broadband providers ($\sigma = 3.65$) is associated with increases in relative party unity of 0.011, 0.015, and 0.022 across the three samples, meaningful within-

Figure 8: Marginal Effect of Broadband on District Representational Outcomes



Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from OLS regression models specified with vector of control variables, year × party fixed-effects, and member fixed-effects.

member shifts in partisan voting behavior over a single electoral cycle. Consistent with H5, Figure 7 finds no evidence of a broadband effect on ideological extremity as measured by Nokken-Poole DW-NOMINATE scores, with null estimates of approximately 0.001 across all three samples. This null result provides strong support for H5 and confirms that the mechanism linking broadband expansion to legislative behavior is fundamentally partisan rather than ideological in nature. The primary threat channeled through a more partisan information environment disciplines incumbent loyalty to the party team rather than demanding spatial ideological repositioning, consistent with the conditional party government framework of Aldrich & Rohde (2000) and the partisan account of party unity advanced by Lee (2009). With respect to co-partisan co-sponsorship, we find a marginally significant negative effect on the proportion of co-partisan co-sponsorships among full observations (-0.009^{*}), though this result does not replicate consistently across the sophomore and complete panel samples, suggesting a less robust relationship between broadband and co-sponsorship behavior relative to the party unity findings.

Turning to Figure 8, we find consistent evidence supporting both H6 and H7. With respect to the null expectations of H7, broadband expansion has no significant effect on legislative effectiveness, constituency staffing, or Members' Representational Allowance spending across any of the three panel samples, with small and statistically insignificant marginal effects throughout. This pattern confirms that the behavioral shift associated with broadband expansion is one of representational orientation rather than material

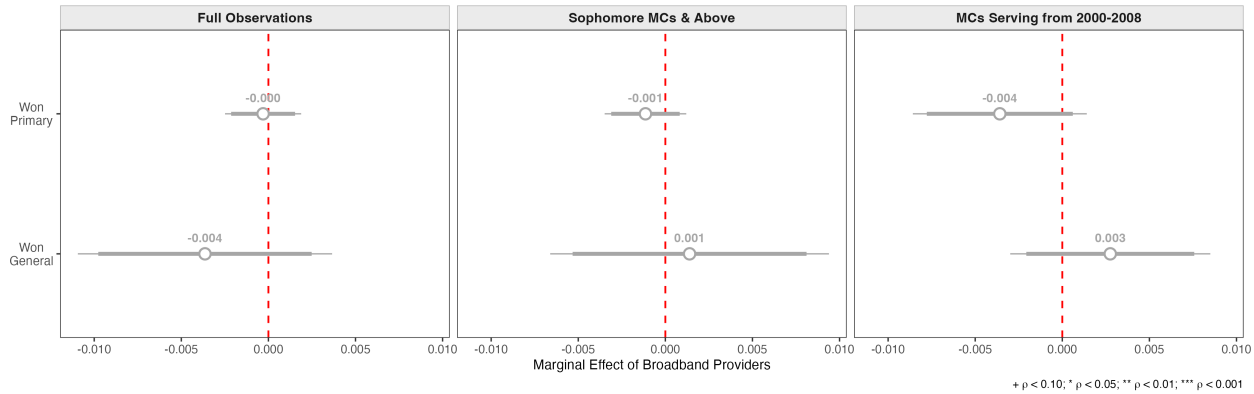
resource allocation: incumbents do not withdraw constituency service resources in response to the partisan primary threat. Consistent with H6, however, Figure 8 shows a significant and consistent negative effect of broadband expansion on the probability that a member adopts a district-focused rather than party or policy focused legislative style. Across the full, sophomore and above, and complete panel samples, one additional broadband provider reduces the probability of a district-focused representational style by 1.7%, 1.4%, and 2.0%, respectively, significant at conventional levels across all three samples. Decomposing this effect, the shift away from district-focused style appears driven primarily by movement toward party-focused rather than policy-focused representation, with broadband showing marginally significant negative effects on district versus party-focused style among the sophomore and complete panel samples while the district versus policy-focused comparison yields null results throughout. Together, the results of Figures Figure 7 and Figure 8 provide strong and theoretically coherent support for the mechanism we theorize: broadband expansion motivates greater partisan loyalty in roll-call voting and a reorientation of representational style away from the geographic constituency and toward the partisan collective, effects that are partisan rather than ideological in character and concentrated in representational orientation rather than material service delivery.

4.3 Anticipatory Adaptation: Electoral Outcomes

The final set of results evaluates H8, the anticipatory adaptation prediction that successful behavioral repositioning by incumbents facing a more partisan primary electorate should neutralize the electoral consequences of broadband expansion. Figure 9 presents the marginal effects of one additional broadband provider on the probability of winning primary renomination and winning the general election across our three panel samples.

Consistent with the predictions of H8, Figure 9 finds no evidence that broadband expansion affects incumbent electoral outcomes across any panel sample or electoral stage. Marginal effects on primary renomination are negligible and statistically indistinguishable from zero, ranging from -0.000 to -0.004 across the three panel samples. Similarly, the effects of broadband on general election success are small, signed inconsistently, and nowhere approaching conventional levels of statistical significance, ranging from

Figure 9: Marginal Effect of Broadband on Primary & General Election Success



Note: Figure shows average marginal effect of one additional broadband provider on outcome variables across three sample restrictions. Thick (thin) lines articulate 90% (95%) confidence intervals calculated using member clustered robust standard errors. Post-estimated marginal effects derived from linear probability models specified with vector of control variables, year × party fixed-effects, and member fixed-effects.

-0.004 to $+0.003$. These null results hold across all three panel samples, providing consistent evidence that broadband expansion does not translate into electoral vulnerability for incumbents.

Taken together, the results of Sections 4.1 through 4.3 present a theoretically coherent empirical portrait of how broadband internet expansion shaped the political behavior of members of Congress during the 2002 to 2008 period. Broadband expansion motivated voluntary retirement and co-partisan primary opposition, drove incumbents toward greater partisan loyalty in roll-call voting and a more partisan representational style, and yet left incumbent electoral outcomes unchanged. This pattern is precisely what the anticipatory adaptation framework of Arnold (1990) predicts: incumbents who perceive a credible primary threat respond by repositioning their behavior toward the preferences of the primary electorate, successfully neutralizing the threat before it produces an electoral consequence. The behavioral disciplining documented in the mechanism tests, greater party unity and reduced district-focused style, is not epiphenomenal but constitutes the strategic response that sustains the null electoral result. Broadband expansion thus operates as a mechanism of partisan disciplining that reshapes the behavioral calculus of incumbents without translating into systematic electoral punishment, suggesting that the primary threat channel operates primarily through anticipatory adaptation rather than realized electoral losses.

5 Discussion: High Information Environments & Demand for Partisan Polarization

> "The internet democracy we have today simplifies too much. And most of the issues that we deal with aren't black and white. The internet democracy puts a premium on instant, quick, certain reactions and in that way makes it difficult to govern."
>
> U.S. Senator Lamar Alexander (R-TN)

Speaking to an audience at his law school alma mater, retiring U.S. Senator Lamar Alexander (R-TN) outlined a concern that internety democracy "puts a premium on instant, quick, certain reactions" that makes it "difficult to govern." [8] This assertion speaks to our findings and frames a paradoxical implication about the expansion of the information environment on the state of American democracy. On the one hand, the expansion of broadband internet during the 2000s gave constituents more critical information about their elected agents that is necessary towards fulfilling their roles as principals guarding against agency loss against unfaithful agents. However, our findings suggest that this new information restructured the informational incentives facing members of Congress in a manner that makes partisan loyalty and polarization a rational survival strategy, at the expense of district centered governance. These incentives empower partisan coalitional networks and the policy-seeking activists that reside in them (Bawn et al., 2012) and participate in first-stage primary elections, with broadband facilitating a rise in a primary threat that legislators must adjust to in an increasingly nationalized and party-centered landscape. Indeed, our work provides a theoretical account as to *why* this happened: changes to the information environment generated a credible co-partisan primary flank, and incumbents responded with anticipatory behavioral adaptation that prioritized partisan signaling over constituency service. The "simplification" Senator Alexander laments is not incidental to the information environment but a predictable consequence of it, as the primary electorate most mobilized by broadband expansion demanded the kind of clear partisan

[8] NYU School of Law, A Q&A with Lamar Alexander '65. https://www.law.nyu.edu/news/lamar-alexander-public-service-education

signals that reward party loyalty and punishes bipartisan compromise.

In doing so, our work speaks to an unresolved puzzle in the primary elections literature. While prior work documents the rise of primary competition (Boatright, 2013; Porter & Treul, 2024) and the association between primaries and legislative polarization (Hirano et al., 2010; Theriault, 2008), less is understood as to what specifically drives these patterns in the first place. Our within-member design identifies the expansion of broadband internet as a specific exogenous source of primary discipline, while our finding that the mechanism is *partisan* rather than *ideological* refines the conventional account of how primaries drive polarization. It is not that incumbents become more extreme in response to primary threats, it is that they become more loyal, a distinction with important implications for how we understand the sources of partisan discipline in Congress. While we concur with prior work that legislators "die in their ideological boots" (**?**), we contend broadband expansion plays a vital role in ensuring members toe the party line as collective parties respond to an era of perpetual electoral positioning during an era of insecure majorities (Lee, 2009, 2016; Algara, 2024). Congruently, our anticipatory adaptation result suggests that incumbents successfully neutralize primary threats through behavioral repositioning before a challenger materializes, helping explain why primary renomination rates remain so high even as primary competition intensifies, a pattern that puzzles scholars and practitioners alike (Rackey & Thorning, 2024).

We hope that this work highlights important questions that merit attention in future research on how information environments can alter the electoral incentives of members of Congress and subsequent representational responses. First, our findings are specific to the 2002 to 2008 broadband diffusion window, before social media platforms reshaped the partisan information environment in ways that may have amplified or attenuated the mechanisms we identify. Whether the primary threat channel operates similarly in the social media era, where the organizational infrastructure of partisan accountability has become far more decentralized and reactive, remains an open question. Second, while our theoretical framework emphasizes the role of broadband in mobilizing engaged partisans into the primary electorate, our design identifies this mechanism indirectly through incumbent behavior rather than directly observing changes in primary electorate composition. A more complete account of how the information environment disciplines legislators would examine whether broadband expansion changed *who* participates in primary elections,

not just how incumbents respond to the threat of primary accountability, providing direct evidence for the demand-side mechanism our theory proposes. Lastly, our work raises the broader question of whether the demand for partisan polarization that the information environment creates is a permanent feature of the contemporary legislative environment or one that could be moderated by changes in the informational infrastructure facing constituents and the primary electorates that discipline their representatives.

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